```
Server:
Transfer-Encoding: chunked
```

#A The first requests succeed while the rate limit is not exceeded.
#B Once the rate limit is exceeded, requests are rejected with a 429 status code.

By returning a 429 response immediately, you can limit the amount of work that your API is performing to the bare minimum, allowing it to use those resources for serving the requests that it can handle. The rate limit should always be set below what you think your servers can handle, to give some wiggle room.

> **MINI-PROJECT** The current implementation limits the overall rate of requests from all clients. Adapt the implementation to instead use a separate `RateLimiter` for each client IP address in a `Map`. You can use `request.ip()` to get the client IP address from Spark.

EXERCISES

1. Which one of the following statements is true about rate-limiting?

a) Rate-limiting should occur after access control.

b) Rate-limiting stops all denial of service attacks.

c) Rate-limiting should be enforced as early as possible.

d) Rate-limiting is only needed for APIs that have a lot of clients.